

Question		Solution	Marks
6	(a)	any two points from: • (maximum) kinetic energy of electrons is independent of intensity • maximum kinetic energy of electrons depends on frequency • no time delay (between illumination and emission)	B2
	(b)	(i) (for $E_{\text{MAX}} = 0$,) $1 / \lambda_0 = 1.93 \times 10^6 \text{ (m}^{-1}\text{)}$ $f_0 = 3.00 \times 10^8 \times 1.93 \times 10^6$ $= 5.8 \times 10^{14} \text{ Hz}$	C1 A1
		$hc / \lambda = \Phi + E_{\text{MAX}}$ $hc = \text{gradient}$ $\text{gradient} = \text{e.g. } [(0.40 - 0.20) \times 1.60 \times 10^{-19}] / [(2.25 - 2.09) \times 10^6]$ (working needed) $(= 2.0 \times 10^{-25})$ $h = (2.0 \times 10^{-25}) / (3.00 \times 10^8) = 6.7 \times 10^{-34} \text{ J s (both working and answer needed)}$	C1 C1 M1 A1
		straight line with same gradient as the original straight line with x-axis intercept greater than $1.93 \times 10^6 \text{ m}^{-1}$	B1 B1

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7	(a)	(i)	q and P are constants hence a is proportional to x	B1